

Supplementary Material

Additional Evaluation Samples

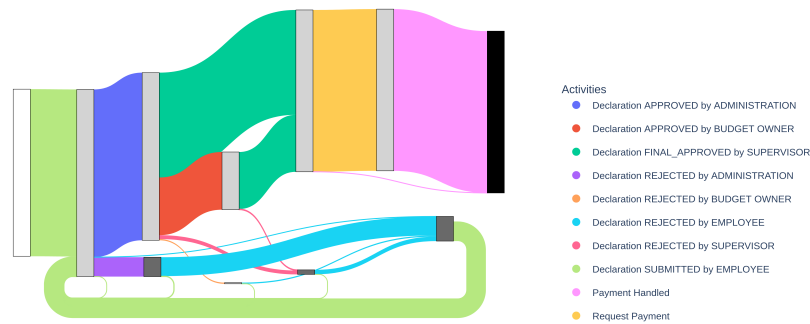


Fig. 9: Travel reimbursement approval process from the *Domestic Declarations* log, filtered for cases in 2018 because 2017 and 2018 have different activity schemas; last activity abstraction visualization: The Sankey shows knockout structure embedded in a rework structure (employee resubmission) including exceptional paths.

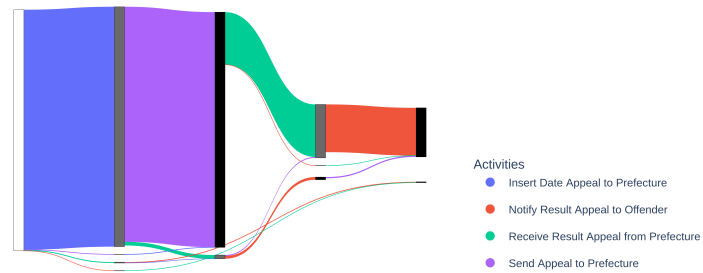


Fig. 10: The prefecture appeal subprocess of the *Road Fines* log in set abstraction: Strong leakage after *Send Appeal* activity with few exceptional paths with unexpected ordering.

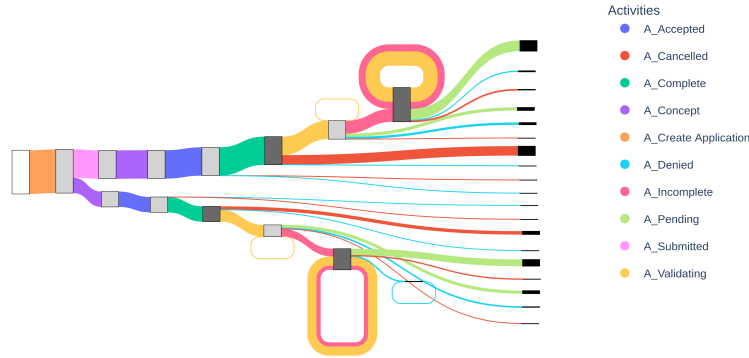


Fig. 11: The application object lifecycle from the BPIC17 loan approval process in set abstraction: Rework is clearly visible as self-loops containing *A_Incomplete* and *A_Validating*. We can also see that the relative extent of rework is roughly the same independent from whether *A_Submitted* happened before or not.

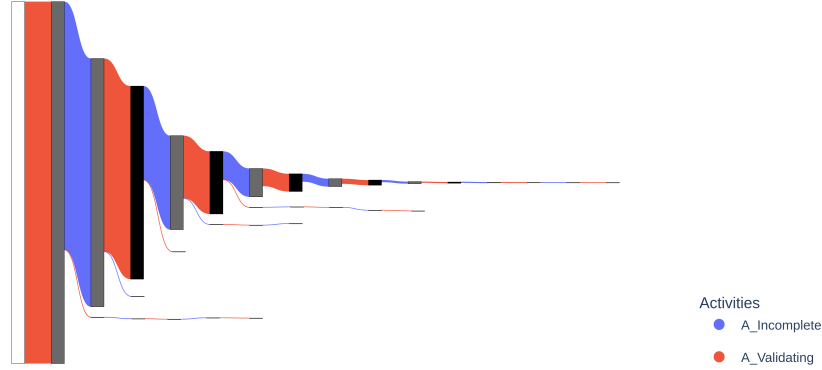


Fig. 12: Rework analysis for the application object of the loan approval process from the BPCI17 log: The sequence abstraction (tree view) shows the number of cases with specific number of repetitions in the same case.

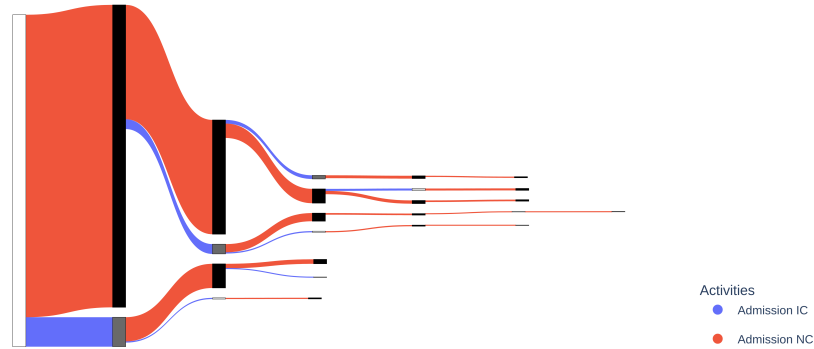


Fig. 13: Rework analysis for the administrative subprocess of the Sepsis log: The sequence abstraction (tree view) shows the number of cases with specific number of repetitions in the same case.

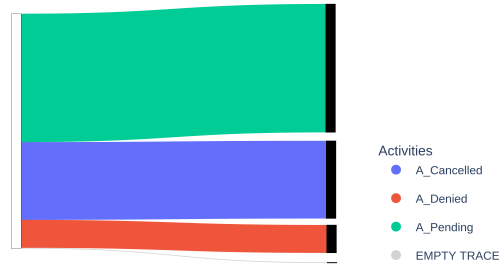


Fig. 14: Outcome distribution of BPIC17 loan applications: loan approved (*A_Pending*), loan denied (*A_Denied*) and application canceled by customer (*A_Cancelled*). A negligible number of applications has not reached an outcome (empty trace). All outcomes are mutually exclusive.

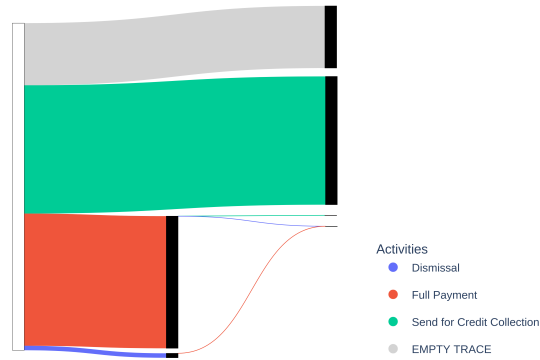


Fig. 15: Outcome distribution for the *Road Fines* log after two activity unfoldings: Payment is unfolded into partial payment and full payment and appeals are unfolded into dismissals and unsuccessful appeals. While we see very few traces that have more than one outcome, there is a large number of traces with no outcome defined.

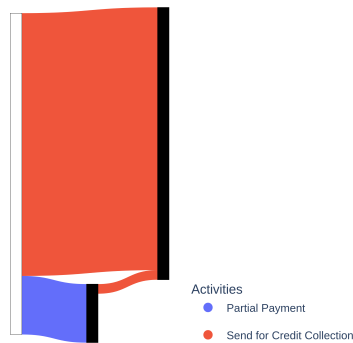


Fig. 16: Prediction of credit collection from presence of a first partial payment: Only one fifth of traces that have a partial payment end up in credit collection (last activity abstraction, *Road Fines* log with activity unfolding).

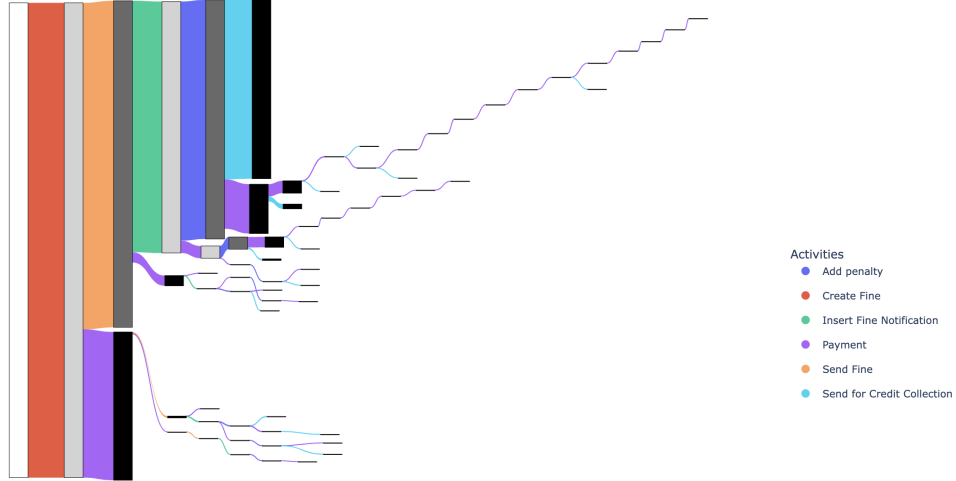


Fig. 17: The prefix tree (sequence abstraction) of the fine subprocess of the unaltered *Road Fines* log. Variation arises only with respect to when and how many *Payment* events occur. Most repeated payments happen in the context when a fine letter has been sent.

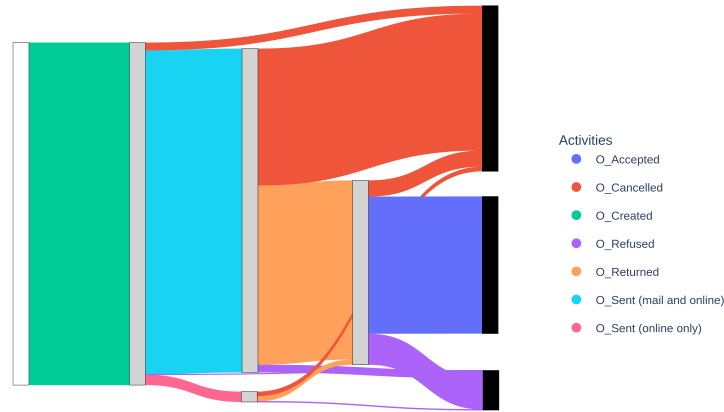


Fig. 18: The last-activity abstraction of the sub-traced BPIC17 log, which isolates the lifecycle of an offer object. Knockout-structure can be seen with the two knockout outcomes, viz. *O_Cancelled* and *O_Refused*. This subprocess is very regular with no exceptional or unexpected sequences. End nodes represent the outcome distribution. We also see the distribution of how many offers are sent online only vs. mail and online.

Results from commercial tool survey

The following show the results from using our test samples detailed in the main paper on Fluxicon Disco, Celonis, and Apromore.

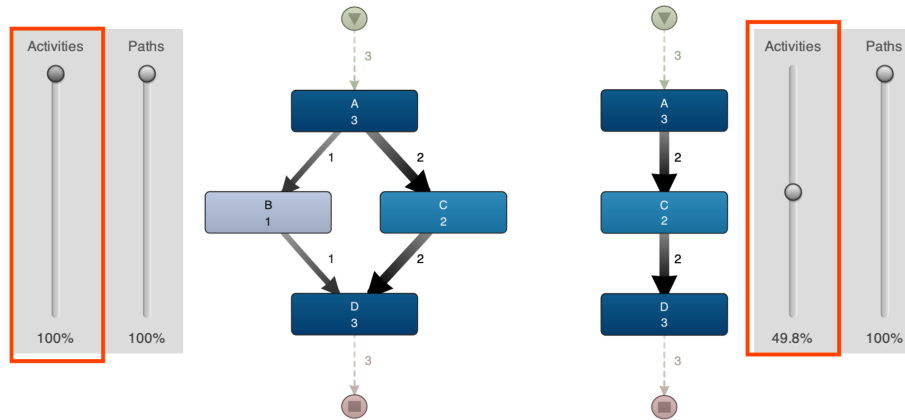


Fig. 19: DFG exhibiting *discontinuity*. Produced with *Fluxicon Disco* with the log *ABD, ACD, ACD* and filter setting activities 49.8% and paths 100%, with changed filters highlighted in red in the figure.

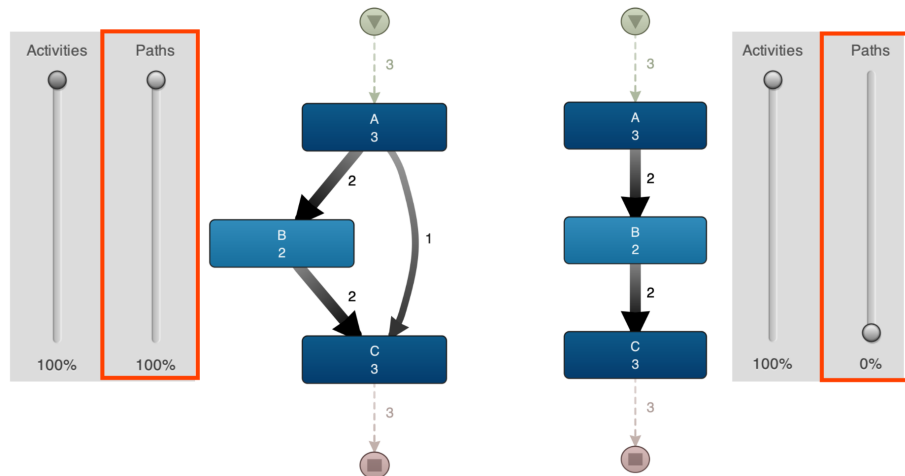


Fig. 20: DFG exhibiting *discontinuity*. Produced with *Fluxicon Disco* with the log *ABC, ABC, AC* and filter setting activities 100% and paths 0%, with changed filters highlighted in red in the figure.

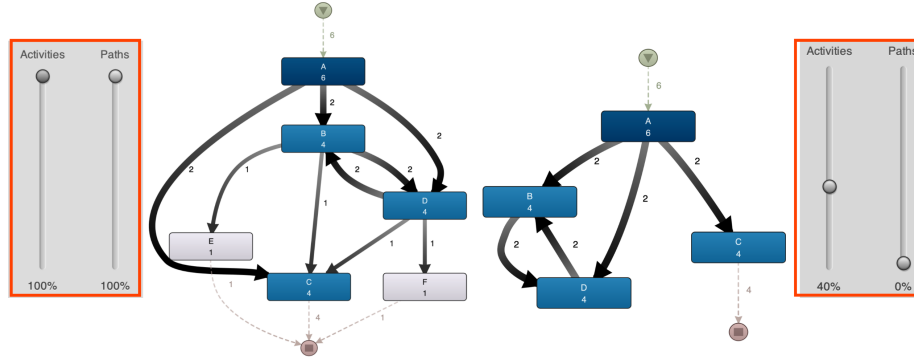


Fig. 21: DFG exhibiting *DFG unsoundness* because from activity *D*, the end node cannot be reached. Produced with *Fluxicon Disco* with the log *AC, AC, ABDC, ADBE, ABDF, ADBC* and filter setting *Activities: 40%* and *Paths: 0%*, with changed filters highlighted in red in the figure.

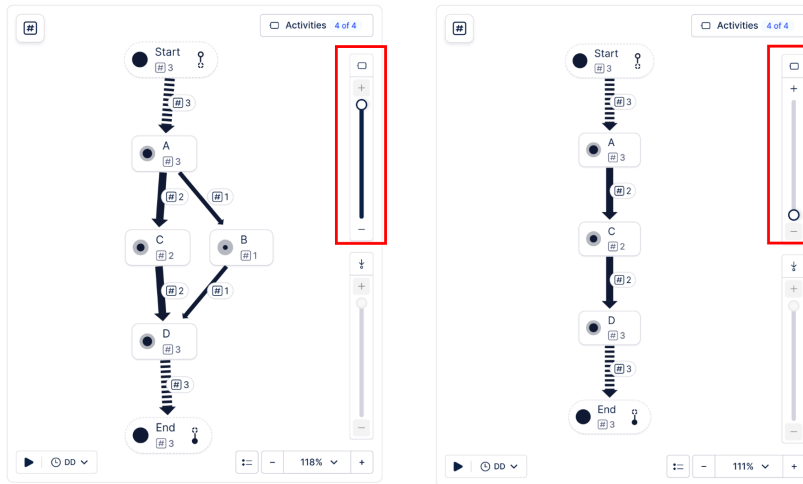


Fig. 22: DFG exhibiting discontinuity and non-linearity. Produced with *Celonis* with the log *ABD, ACD, ACD* and filter setting (changes) as highlighted in red in the figure.

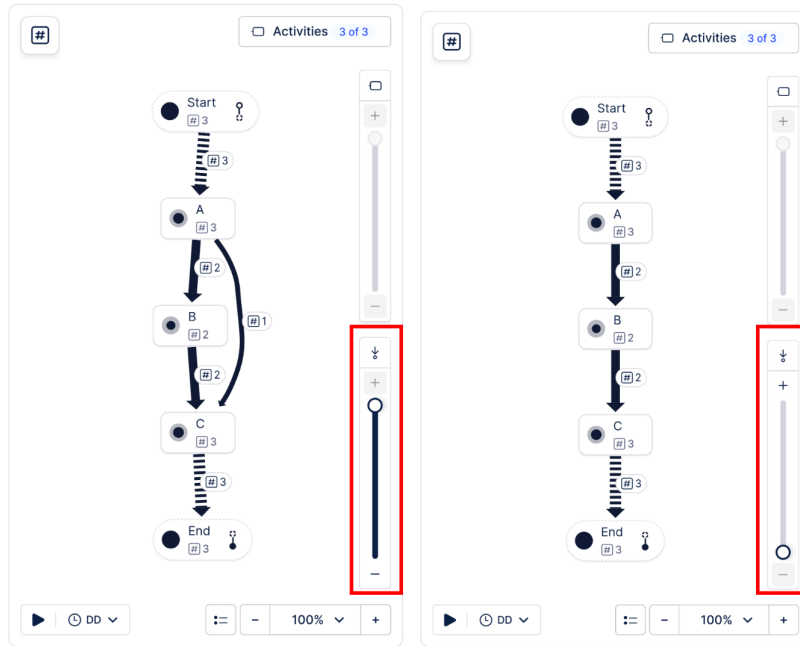


Fig. 23: DFG exhibiting discontinuity and non-linearity. Produced with *Celonis* with the log ABC, ABC, AC and filter setting (changes) as highlighted in red in the figure.

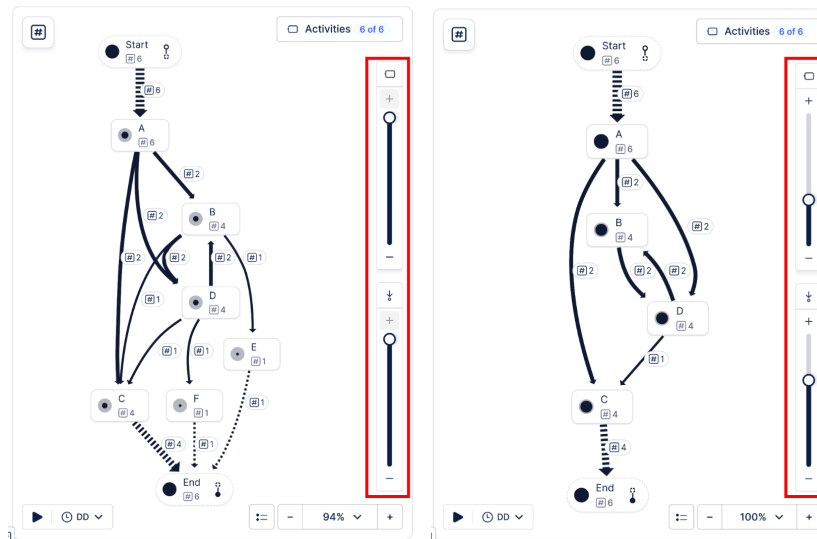


Fig. 24: DFG exhibiting lack of DFG unsoundness (cf. Supplement, Fig. 21).. Produced with *Celonis* with the log $AC, AC, ABDC, ADBE, ABDF, ADBC$ and filter setting (changes) as highlighted in red in the figure.

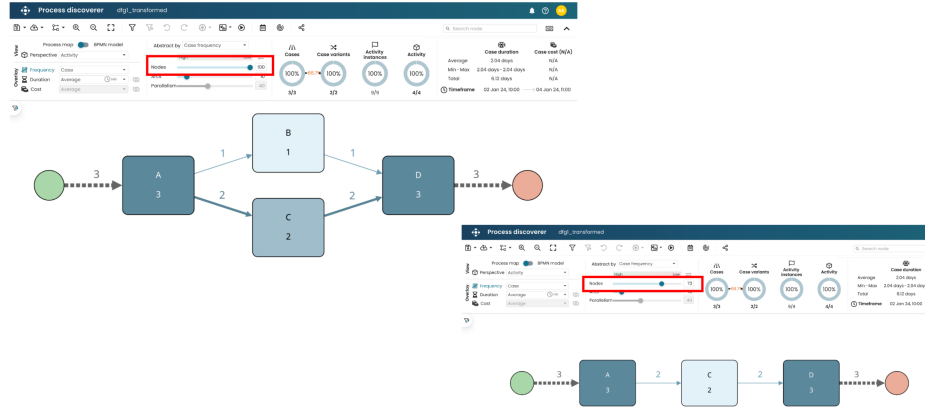


Fig. 25: DFG exhibiting discontinuity and non-linearity. Produced with *Apromore* with the log ABD, ACD, ACD and filter setting nodes 73% and arcs 10%, with changed filters highlighted in red in the figure.

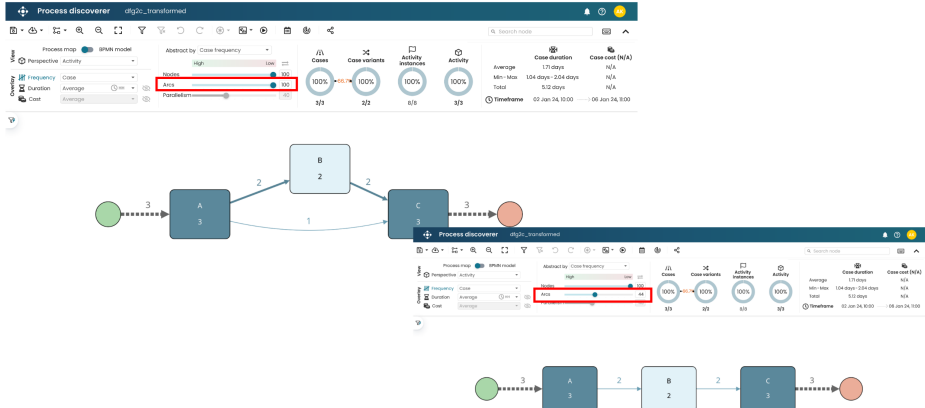


Fig. 26: DFG exhibiting discontinuity and non-linearity. Produced with *Apromore* with the log ABC, ABC, AC and filter setting nodes 100% and arcs 44%, with changed filters highlighted in red in the figure.

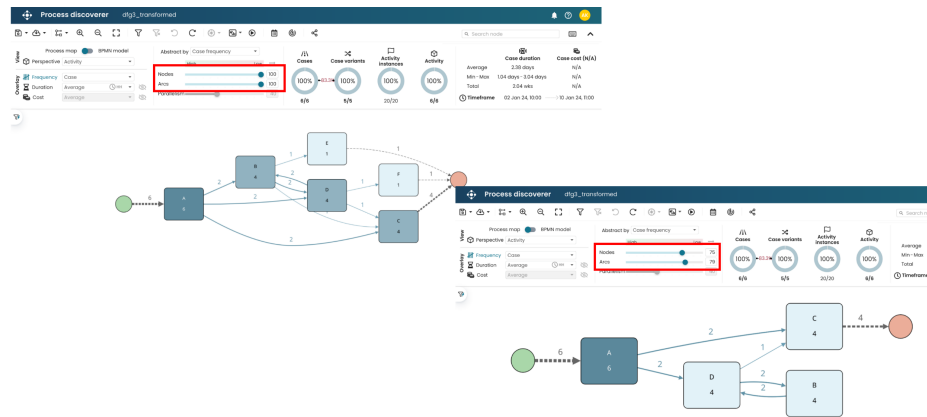


Fig. 27: DFG exhibiting lack of DFG unsoundness (cf. Supplement, Fig. 21). Produced with *Apromore* with the log *AC, AC, ABDC, ADBE, ABDF, ADBC* and filter setting nodes 75% and arcs 79%, with changed filters highlighted in red in the figure.